

1. Determine the points of maximum and minimum of the function $f(x, y) := \int_0^{x+y^2} e^{-t^2} dt$ subject to $2x^2 + y^2 \leq 1$.

The several variable version of the extremum value theorem (from your first year calculus) states that a continuous functions achieves its maximum and minimum values on a closed (including its boundary) and bounded domain. So, we expect the function f to achieve its maximum and minimum values on the elliptical disc $D := \{(x, y) \in \mathbb{R}^2: 2x^2 + y^2 \leq 1\}$.

- (i) Any point of minimum or maximum of the function f in the interior of the elliptical disc i.e., $2x^2 + y^2 < 1$, is a critical point. So, we first find the critical points of the function f which lie in the interior $2x^2 + y^2 < 1$.

By a combination of the fundamental theorem of calculus and the chain rule, we get

$$f_x(x, y) = e^{-(x+y^2)^2} \quad \text{and} \quad f_y(x, y) = 2ye^{-(x+y^2)^2}.$$

The function f has no critical points since $f_x(x, y) \neq 0$.

- (ii) Now, we find the points of maximum and minimum of the function f on the ellipse $2x^2 + y^2 = 1$ using the method of the Lagrange multiplier as follows: we seek x, y, λ such that

$$\begin{cases} e^{-(x+y^2)^2} = 4\lambda x, \\ 2ye^{-(x+y^2)^2} = 2\lambda y \\ 2x^2 + y^2 = 1 \end{cases} \Leftrightarrow \begin{cases} e^{-(x+y^2)^2} = 4\lambda x, \\ y[\lambda - e^{-(x+y^2)^2}] = 0 \\ 2x^2 + y^2 = 1 \end{cases}$$

Notice that $\lambda \neq 0$ from the first equation and from the second equation, we get $y = 0$ or $\lambda - e^{-(x+y^2)^2} = 0$.

- If $y = 0$, we get from the third equation that $x = \pm\sqrt{\frac{1}{2}}$ and therefore, we obtain the points

$$\left(-\sqrt{\frac{1}{2}}, 0 \right) \quad \text{and} \quad \left(\sqrt{\frac{1}{2}}, 0 \right).$$

- Now assume that $y \neq 0$, that is $\lambda = e^{-(x+y^2)^2}$, from the second equation. Thus, from the first equation we get that $x = \frac{1}{4}$ which substituted into the third equation gives $y = \pm\sqrt{\frac{7}{8}}$. We therefore obtain the points

$$\left(\frac{1}{4}, -\sqrt{\frac{7}{8}} \right) \quad \text{and} \quad \left(\frac{1}{4}, \sqrt{\frac{7}{8}} \right).$$

(iii) Now, we evaluate the function f at the points found in (ii) and compare to get

$$\left\{ \begin{array}{l} f\left(-\sqrt{\frac{1}{2}}, 0\right) = \int_0^{-\sqrt{\frac{1}{2}}} e^{-t^2} dt = -\int_{-\sqrt{\frac{1}{2}}}^0 e^{-t^2} dt < 0, \quad f\left(\sqrt{\frac{1}{2}}, 0\right) = \int_0^{\sqrt{\frac{1}{2}}} e^{-t^2} dt > 0, \\ f\left(\frac{1}{4}, -\sqrt{\frac{7}{8}}\right) = f\left(\frac{1}{4}, \sqrt{\frac{7}{8}}\right) = \int_0^{\frac{1}{4} + \frac{7}{8}} e^{-t^2} dt = \int_0^{\frac{9}{8}} e^{-t^2} dt, \\ f\left(-\sqrt{\frac{1}{2}}, 0\right) < 0 < f\left(\sqrt{\frac{1}{2}}, 0\right) = \int_0^{\sqrt{\frac{1}{2}}} e^{-t^2} dt < \int_0^{\frac{9}{8}} e^{-t^2} dt = f\left(\frac{1}{4}, -\sqrt{\frac{7}{8}}\right) = f\left(\frac{1}{4}, \sqrt{\frac{7}{8}}\right). \end{array} \right.$$

Therefore, $\left(-\sqrt{\frac{1}{2}}, 0\right)$ is the point of minimum while $\left(\frac{1}{4}, -\sqrt{\frac{7}{8}}\right)$ and $\left(\frac{1}{4}, \sqrt{\frac{7}{8}}\right)$ are the points of maximum.

2. Evaluation of double integrals over rectangular domains:

Let $D := [a, b] \times [c, d]$ and $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuous function. Then the double integral of the function f over the domain D is evaluated as follows:

$$\boxed{\iint_D f(x, y) dx dy = \int_a^b \left(\int_c^d f(x, y) dy \right) dx = \int_c^d \left(\int_a^b f(x, y) dx \right) dy} \quad (*)$$

Evaluate the double integrals:

(a) $\iint_D \sin(x - y) dx dy$ where $D := \left[0, \frac{\pi}{2}\right] \times [0, \pi]$.

By successive integration, we find that

$$\begin{aligned} \iint_D \sin(x - y) dx dy &= \int_0^{\frac{\pi}{2}} \int_0^{\pi} \sin(x - y) dy dx = \int_0^{\frac{\pi}{2}} \left[\cos(x - y) \right]_0^{\pi} dx \\ &= \int_0^{\frac{\pi}{2}} (\cos(x - \pi) - \cos x) dx = \left[\sin(x - \pi) - \sin x \right]_0^{\frac{\pi}{2}} = -2. \end{aligned}$$

(b) $\iint_D x \arctan\left(\frac{1}{x}\right) \cos^3(y) dx dy$ where $D := [1, \sqrt{3}] \times \left[0, \frac{\pi}{2}\right]$.

Hint: $\cos^3(\theta) = (\cos^2 \theta) \cos \theta = (1 - \sin^2 \theta) \cos \theta = \cos \theta - \sin^2 \theta \cos \theta$.

By successive integration in (*), we find that

$$\begin{aligned} \iint_D x \arctan\left(\frac{1}{x}\right) \cos^3(y) dx dy &= \int_1^{\sqrt{3}} \int_0^{\frac{\pi}{2}} x \arctan\left(\frac{1}{x}\right) \cos^3(y) dy dx \\ &= \int_1^{\sqrt{3}} x \arctan\left(\frac{1}{x}\right) \int_0^{\frac{\pi}{2}} \cos^3(y) dy dx = \left(\int_1^{\sqrt{3}} x \arctan\left(\frac{1}{x}\right) dx \right) \left(\int_0^{\frac{\pi}{2}} \cos^3(y) dy \right) \end{aligned}$$

By integration by parts, we find that

$$\begin{aligned}
\int_1^{\sqrt{3}} x \arctan\left(\frac{1}{x}\right) dx &= \left[\frac{1}{2} x^2 \arctan\left(\frac{1}{x}\right) \right]_1^{\sqrt{3}} - \int_1^{\sqrt{3}} \frac{1}{2} x^2 \frac{-\frac{1}{x^2}}{1 + \frac{1}{x^2}} dx \\
&= \left[\frac{1}{2} x^2 \arctan\left(\frac{1}{x}\right) \right]_1^{\sqrt{3}} + \frac{1}{2} \int_1^{\sqrt{3}} \frac{x^2}{x^2 + 1} dx \\
&= \left[\frac{1}{2} x^2 \arctan\left(\frac{1}{x}\right) \right]_1^{\sqrt{3}} + \frac{1}{2} \int_1^{\sqrt{3}} \left(1 - \frac{1}{x^2 + 1} \right) dx = \left[\frac{1}{2} x^2 \arctan\left(\frac{1}{x}\right) + \frac{1}{2} x - \frac{1}{2} \arctan x \right]_1^{\sqrt{3}} \\
&= \left(\frac{3}{2} \arctan\left(\frac{1}{\sqrt{3}}\right) + \frac{1}{2} \sqrt{3} - \frac{1}{2} \arctan(\sqrt{3}) \right) - \left(\frac{1}{2} \arctan(1) + \frac{1}{2} - \frac{1}{2} \arctan(1) \right) \\
&= \frac{3\pi}{12} + \frac{\sqrt{3}}{2} - \frac{\pi}{6} - \frac{1}{2} = \frac{\pi}{12} + \frac{\sqrt{3} - 1}{2}.
\end{aligned}$$

On the other hand, using the hint we obtain

$$\int_0^{\frac{\pi}{2}} \cos^3(y) dy = \int_0^{\frac{\pi}{2}} (\cos y - \sin^2 y \cos y) dy = \left[\sin y - \frac{1}{3} \sin^3 y \right]_0^{\frac{\pi}{2}} = \frac{2}{3}.$$

Therefore, we find that

$$\begin{aligned}
\iint_D x \arctan\left(\frac{1}{x}\right) \cos^3(y) dx dy &= \left(\int_1^{\sqrt{3}} x \arctan\left(\frac{1}{x}\right) dx \right) \left(\int_0^{\frac{\pi}{2}} \cos^3(y) dy \right) \\
&= \left[\frac{\pi}{12} + \frac{\sqrt{3} - 1}{2} \right] \times \frac{2}{3} = \frac{\pi}{18} + \frac{\sqrt{3} - 1}{3}.
\end{aligned}$$

(c) $\iint_D \cos^2(x + y) dx dy$ where $D := [0, \frac{\pi}{4}] \times [-\frac{\pi}{4}, \pi]$.

Hint: $\cos(2\theta) = 2 \cos^2 \theta - 1$.

By successive integration in (*) and using the hint we find that

$$\begin{aligned}
\iint_D \cos^2(x + y) dx dy &= \int_{-\frac{\pi}{4}}^{\pi} \int_0^{\frac{\pi}{4}} \cos^2(x + y) dx dy = \int_{-\frac{\pi}{4}}^{\pi} \int_0^{\frac{\pi}{4}} \frac{\cos(2(x + y)) + 1}{2} dx dy \\
&= \frac{1}{2} \int_{-\frac{\pi}{4}}^{\pi} \left[\frac{1}{2} \sin(2x + 2y) + x \right]_0^{\frac{\pi}{4}} dy \\
&= \frac{1}{2} \int_{-\frac{\pi}{4}}^{\pi} \left(\frac{1}{2} \sin\left(\frac{\pi}{2} + 2y\right) + \frac{\pi}{4} - \frac{1}{2} \sin(2y) \right) dy \\
&= \frac{1}{2} \left[-\frac{1}{4} \cos\left(\frac{\pi}{2} + 2y\right) + \frac{\pi}{4} y + \frac{1}{4} \cos(2y) \right]_{-\frac{\pi}{4}}^{\pi} \\
&= \frac{1}{2} \left[\left(0 + \frac{\pi^2}{4} + \frac{1}{4} \right) - \left(-\frac{1}{4} - \frac{\pi^2}{16} + 0 \right) \right] = \frac{5\pi^2 + 8}{32}.
\end{aligned}$$

3. Let R be the region in the first quadrant bounded by the graphs of the parabolas $y = 2 - x^2$, $y = x^2$ and the y -axis.

- (i) The two points of intersection are $(-1, 1)$ and $(1, 1)$
- (ii) Sketch the region R .
- (iii)

$$A(R) = \int_0^1 \left[\int_{x^2}^{2-x^2} dy \right] dx = \int_0^1 (2 - 2x^2) dx = \left[2x - \frac{2}{3}x^3 \right]_0^1 = \frac{4}{3}.$$

Notice that we need not to evaluate!

- (iv) In this case we split the region in two parts.

$$\begin{aligned} A(R) &= \int_0^1 \left[\int_0^{\sqrt{y}} dx \right] dy + \int_1^2 \left[\int_0^{\sqrt{2-y}} dx \right] dy \\ &= \int_0^1 \sqrt{y} dy + \int_1^2 \sqrt{2-y} dy = \left[\frac{2}{3} y^{3/2} \right]_0^1 + \left[-\frac{2}{3} (2-y)^{3/2} \right]_1^2 \\ &= \frac{2}{3} + \frac{2}{3} = \frac{4}{3} \end{aligned}$$

Notice that we need not to evaluate!

- (v) Use either (iii) or (iv) to calculate the area of R .
- (vi) By symmetry, $A(R') = 2A(R) = \frac{8}{3}$

4. (a) (i) Can you integrate $\sin(x^2)$ with respect to x ?

No, that integral cannot be evaluated through any of the techniques of integration (by parts, substitution, trigonometric substitution, partial fractions) from first year calculus.

- (ii) Sketch the region of integration which is the region in the first quadrant bounded by the parabola $x = y^2$, the x -axis and the line $x = \sqrt{\pi}$! Since we cannot integrate starting from the integration w.r.t to x , we change the order of integration to start first with the integration w.r.t. y , according to the second Fubini's theorem and get

$$\begin{aligned} \int_0^{\pi^{1/4}} \int_{y^2}^{\sqrt{\pi}} y \sin(x^2) dx dy &= \int_0^{\sqrt{\pi}} \left[\int_0^{\sqrt{x}} y dy \right] \sin(x^2) dx = \frac{1}{2} \int_0^{\sqrt{\pi}} x \sin(x^2) dx \\ &= \left[-\frac{1}{4} \cos(x^2) \right]_0^{\sqrt{\pi}} = \frac{1}{2}. \end{aligned}$$

- (b) Evaluate the double integral

$$\begin{aligned} &\iint_D \frac{x}{(x+y)^2} dx dy \text{ where } D := \{(x, y) \in \mathbb{R}^2: 1 \leq x \leq 2, \quad x^2 \leq y \leq 2x^2\}. \\ &\iint_D \frac{x}{(x+y)^2} dx dy = \int_1^2 \int_{x^2}^{2x^2} \frac{x}{(x+y)^2} dy dx = \int_1^2 \left[-\frac{x}{x+y} \right]_{x^2}^{2x^2} dx \\ &= \int_1^2 \left[\frac{x}{x+x^2} - \frac{x}{x+2x^2} \right] dx = \int_1^2 \left[\frac{1}{x+1} - \frac{1}{1+2x} \right] dx = \left[\ln(1+x) - \frac{1}{2} \ln(1+2x) \right]_1^2 \\ &= (\ln 3 - \frac{1}{2} \ln 5) - (\ln 2 - \frac{1}{2} \ln 3) = \frac{3}{2} \ln 3 - \frac{1}{2} \ln 5 - \ln 2 = \ln \left(\frac{\sqrt{27}}{2\sqrt{5}} \right). \end{aligned}$$